

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Date: _____

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Problem Statement & Background

Problem Statement:

When one GaN transistor switches, its fast voltage swing pushes charge into the gate of the other transistor, the one that is supposed to stay off. That gate rises to **1.65 V, and only 1.4 V is needed to turn it on.** Both devices conduct at once and the supply shorts through them. (That 1.65 V is our device model's; how much the result depends on the model is itself a finding, and it has its own slide.)

Background & Significance:

GaN half-bridges sit inside EV inverters and battery chargers. GaN is chosen because it switches fast, and that speed is exactly what causes this fault.

Existing Solutions:

Gate drivers that turn the device on in steps, clamp the off gate down, adjust the gap between the two switching edges, and hold the gate at -2 V [1]–[4]. Reported: 30.5 % less overshoot, 75 % less turn-off loss [5]–[7].

Limitations / Research Gap:

The reported gains do not separate what the re-tuning is worth. Those gains mix two different things: picking one good setting, and changing the setting while the converter runs. Only the second needs a sensor, a lookup table and a controller.

Aim and Approach

Aim:

To find out how much of a gate driver's benefit needs the driver to re-tune itself while the converter is running, and how much comes from choosing one good setting and leaving it fixed.

Proposed Solution:

A GaN buck converter driven by a **segmented gate driver**: 8 pull-up steps, 8 pull-down steps, an adjustable dead time, a Miller clamp and a -2 V off rail. Every one of them is a setting we can change and measure.

How we approached it:

- 1. Build the converter and check it converts.** 100 V DC in, 48.50 V DC out at 4.869 A: 236.26 W delivered, 97.49 % efficient.
- 2. Recreate the fault.** At the fastest setting the gate that should be off reaches 1.65 V, against a 1.4 V turn-on threshold.
- 3. Fix it one change at a time,** measuring each change on its own run, not all at once.
- 4. Ask whether one fixed setting is enough,** by running the driver at two operating points and seeing whether the best setting moves.

Scope and tools:

In scope: the converter, the driver, its settings, and the FPGA controller. **Tools:** ngspice for every simulation, LTspice to draw the circuit and re-check, Vivado for the FPGA.

System Architecture

The driver is drawn as `kicad/gan_segdrv.kicad_sch` and placed twice in `kicad/gan_buck.kicad_sch` — once per gate.

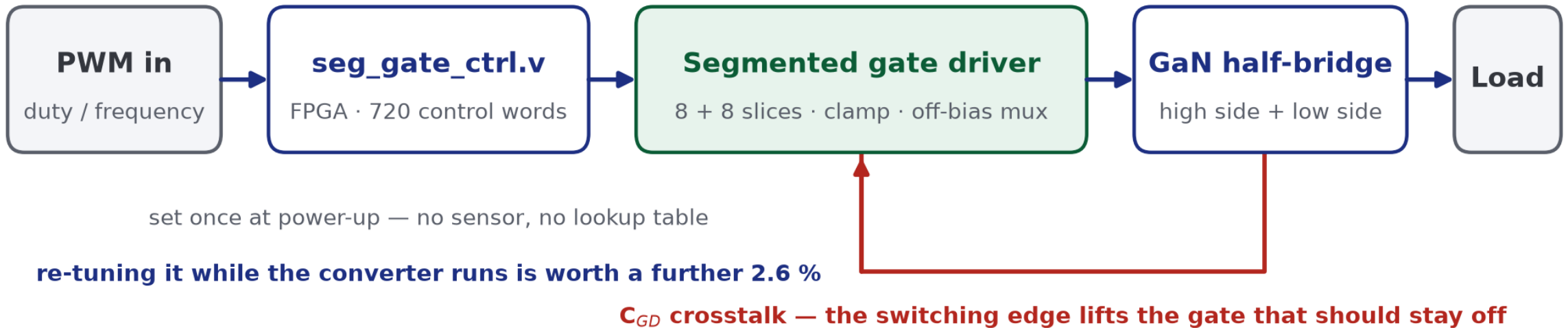


Fig. 1 - Drawn diagram, not simulation output. One signal path: a PWM command, the FPGA controller that holds the control word, the segmented driver that shapes the edge, and the half-bridge it drives. The driver in green is what this project designs; it is drawn once as [kicad/gan_segdrv.kicad_sch](#) and placed twice in [kicad/gan_buck.kicad_sch](#), one per gate. The control word is set at power-up and never changed - re-tuning it while the converter runs is the 2.6 % this project measures. In red, the fault: the switching edge couples through C_{GD} into the gate that should stay off.

Converter Under Simulation

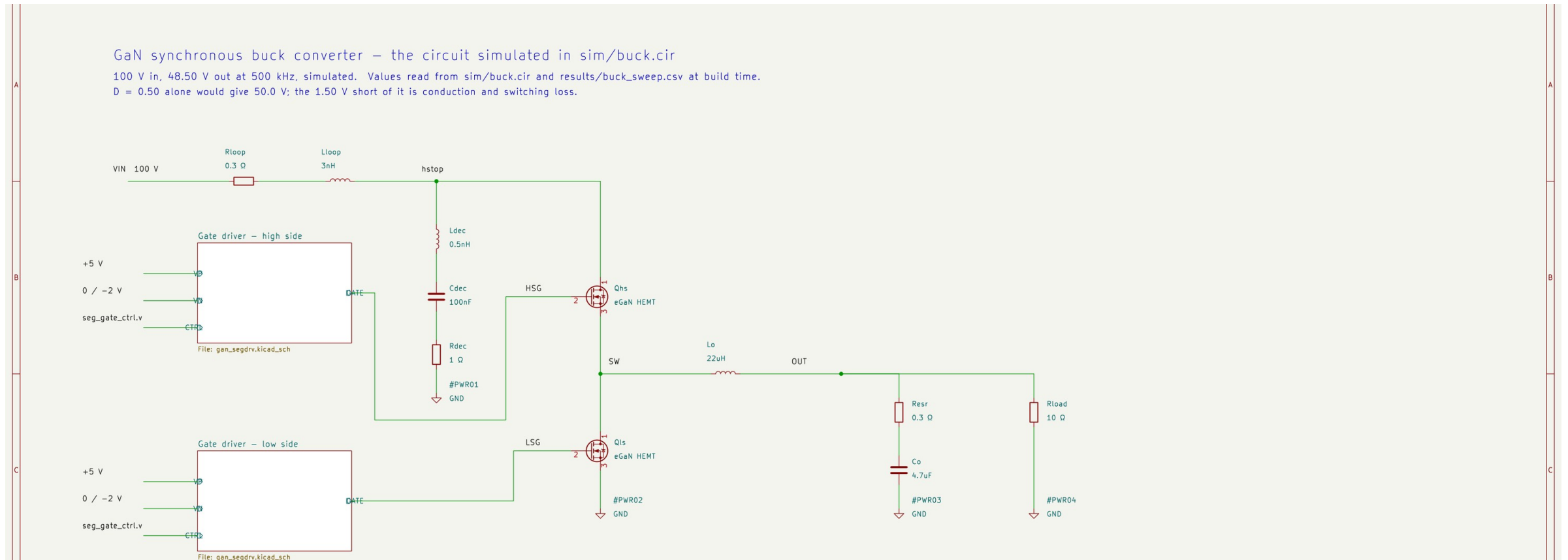


Fig. 2 - Drawn diagram, not simulation output. GaN synchronous buck converter, drawn from sim/buck.cir by [scripts/kicad_schematic.py](#) into [kicad/gan_buck.kicad_sch](#) - every value on it is read out of the netlist at build time. “Buck” means step-down: 100 V in, 48.50 V out. Left to right: the 100 V supply with its power-loop R and L, the damped bus decoupling branch (Ldec, Cdec, Rdec at 1 Ω), the two GaN HEMTs whose gates the segmented drivers drive, the output filter and a 10 Ω load.

Novelty: Three Added Blocks

Same command, same segmented output stage, same power stage. Three blocks added - in green.

	BASE PAPER — Zhang et al., ISPSD 2020	OURS
Control of the driver	one analogue bias resistor, set at fabrication	seg_gate_ctrl.v — digital, re-writable while it runs
Output stage	7 + 7 slices, two stages	8 + 8 slices, thermometer coded
Active Miller clamp	none	always on
Gate off rail	0 V	selectable 0 V / -2 V
Crosstalk margin at 100 V / 10 A	+0.407 V	+2.576 V

Shaded: the three blocks we add. Everything else is theirs, unchanged — same command, same power stage. 6.3× the margin, from one fixed setting.

Fig. 3 - Drawn diagram, not simulation output. The five things that differ, and nothing else. Their column is the base paper as we rebuilt it in [models/zhangdrv.lib](#); ours is [models/segdrv.lib](#). Shaded: the three we add - digital control in place of one fixed resistor, an always-on Miller clamp, and an off rail selectable to -2 V. The command, the power stage and the idea of segmenting the output are all theirs. The last row is what those three buy at the headline corner, read from [results/headtohead.csv](#): +0.407 V becomes +2.576 V, from one fixed setting.

Novelty at Circuit Level

The converter on top; underneath it, what sits behind its two gates in their design and in ours. Both additions are established practice [1]-[4]; the base paper uses neither. What is measured here is what each is worth on this converter.

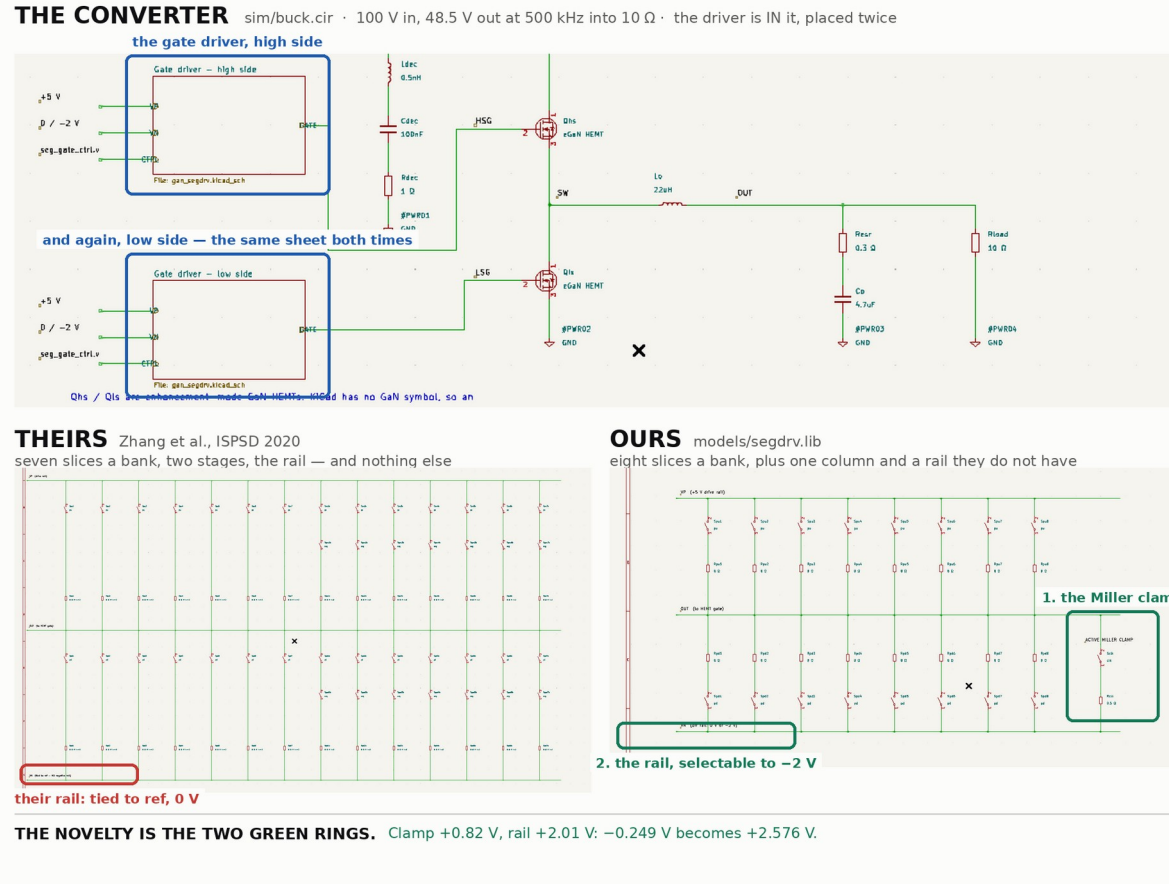


Fig. 4 - Top: [kicad/gan_buck.kicad_sch](#), with the two gates of the half-bridge ringed - HSG and LSG are what a gate driver drives, and everything below is what sits behind them. Bottom left, theirs ([kicad/gan_zhangdrv.kicad_sch](#)): seven slices a bank in two stages, and a rail their own sheet labels “tied to ref - NO negative rail”. Bottom right, ours ([kicad/gan_segdrv.kicad_sch](#)): the same eight-slice output stage, plus the active Miller clamp and an off rail selectable to -2 V. The clamp is worth +0.82 V of crosstalk margin and the rail +2.01 V; together -0.249 V becomes +2.576 V. Annotated screen captures of KiCad 7.0.11, not redrawings, and not a size comparison - the three sheets open at different zoom.

Gate Driver Schematics in KiCad

The two sheets the previous slide crops from - whole, at the same scale, nothing ringed and nothing removed.

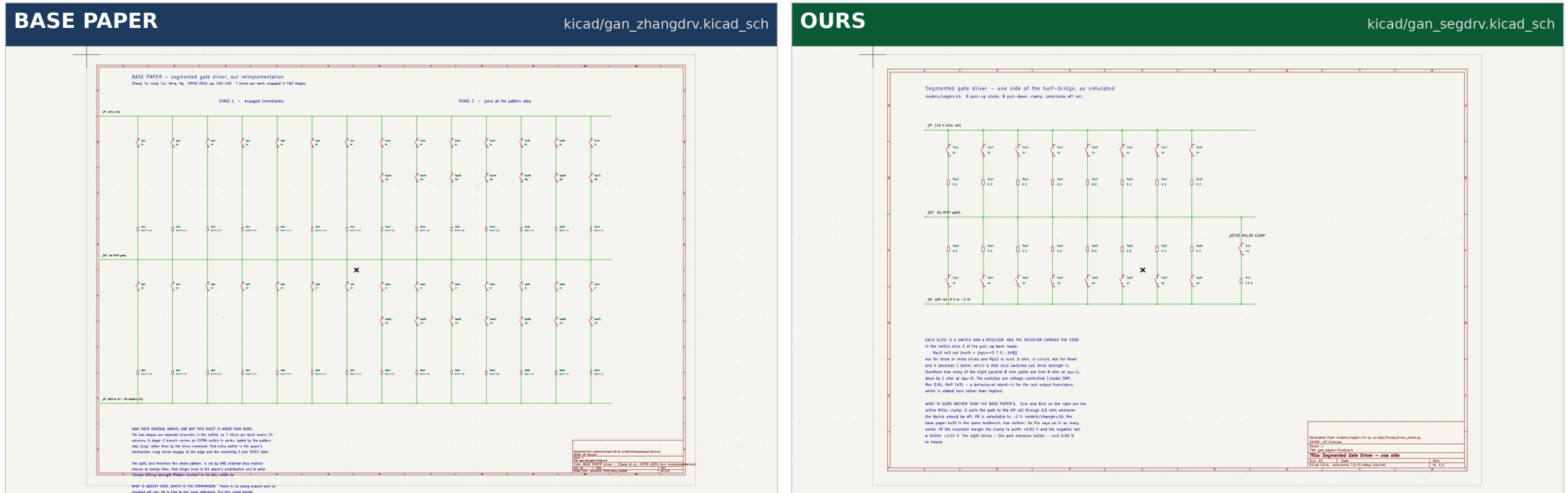


Fig. 5 - Left: [kiCad/gan_zhangdrv.kicad_sch](#), the base paper's driver as we rebuilt it - seven slices a bank in two stages, one bias resistor, and a rail its own sheet labels as tied to reference. Right: [kiCad/gan_segdrv.kicad_sch](#), ours - eight slices a bank, an active Miller clamp and an off rail selectable to -2 V. Both are screen captures of KiCad 7.0.11 taken by [scripts/record_kicad.py](#) from the files named above, not redrawings. The two sheets are different sizes, so this is a comparison of what is on them, not of how large they are.

Crosstalk Mechanism at Device Level

Why a fast edge turns on the device that should be off, and what the driver does about it.

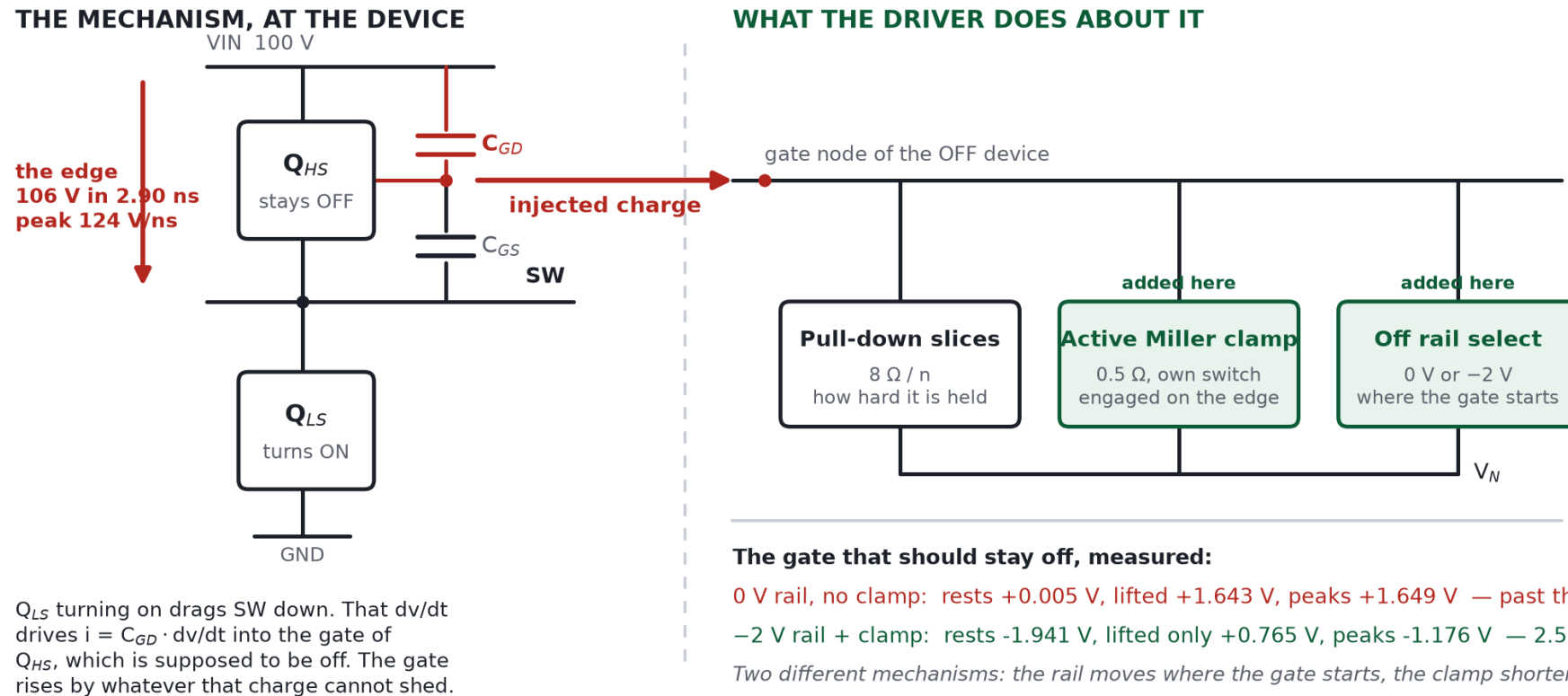


Fig. 6 - Left: the half-bridge. When the low-side device turns on the switch node falls 106 V in 2.90 ns, peaking at 124 V/ns, and that dv/dt drives a current through C_{GD} into the high-side gate. Right: the three parallel paths that current sees at the gate node. The slice count sets how hard the gate is held; the Miller clamp adds a 0.5 ohm path of its own, timed to the other device's edge; the off rail sets where the gate starts from. The two additions are not one mechanism - the rail moves the starting point, the clamp shortens the lift - which is why they add rather than overlap. [scripts/gan_level_figure.py](#), from [results/waveform_anatomy.txt](#).

Implementation: Converter Schematic

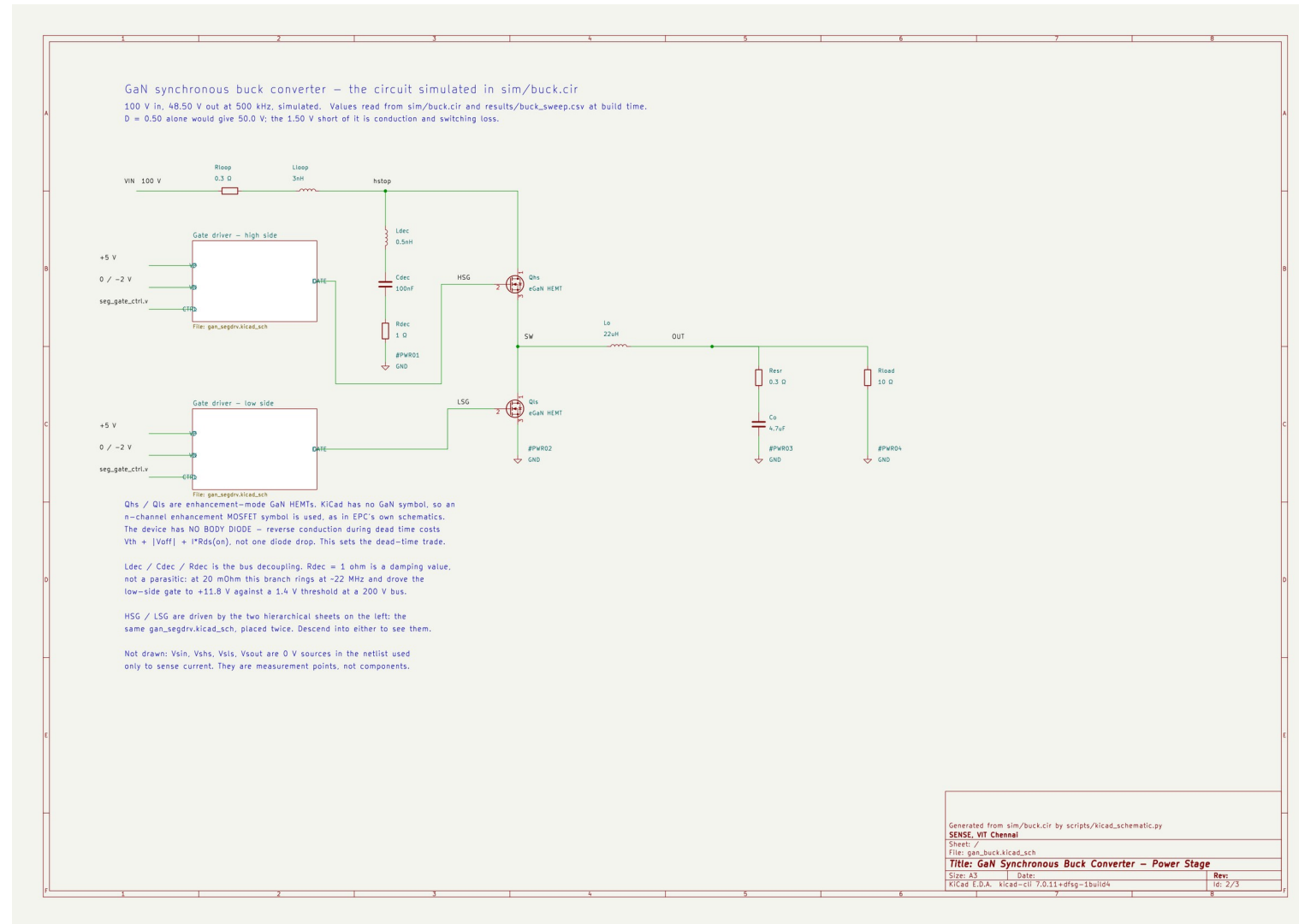


Fig. 7 - `kicad/gan_buck.kicad_sch`, the root sheet, plotted by KiCad 7.0.11. Drawn from [sim/buck.cir](#), so every value on it is the netlist's. The two gate drivers are hierarchical sub-sheets of this page.

Implementation: Proposed Gate Driver

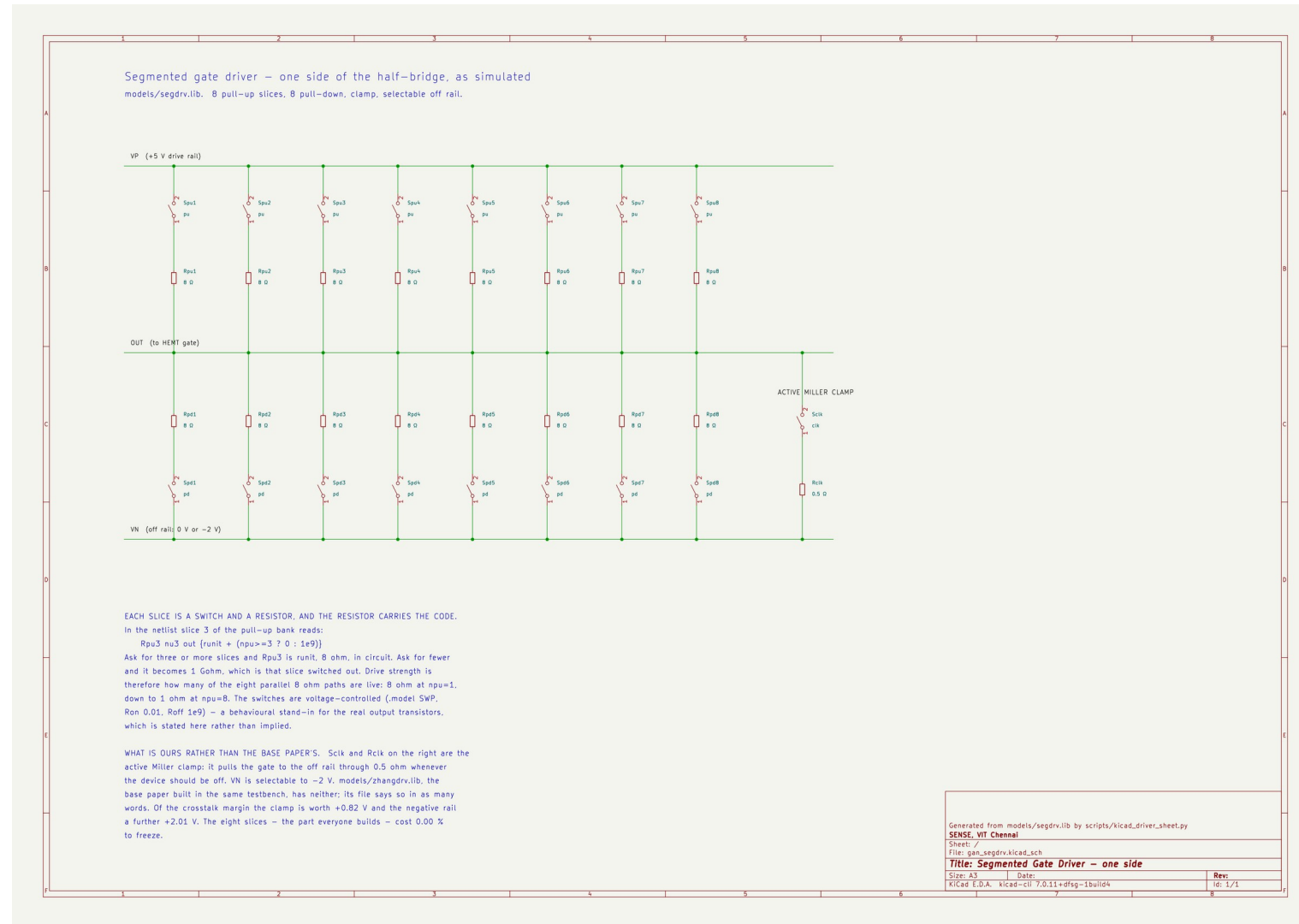


Fig. 8 - [kicad/gan_segdrv.kicad_sch](#). Eight pull-up slices, eight pull-down, the active Miller clamp on the right, and the off rail selectable to -2 V. Each slice is one switch and one resistor; the resistor carries the control word.

Implementation: Base Paper Gate Driver

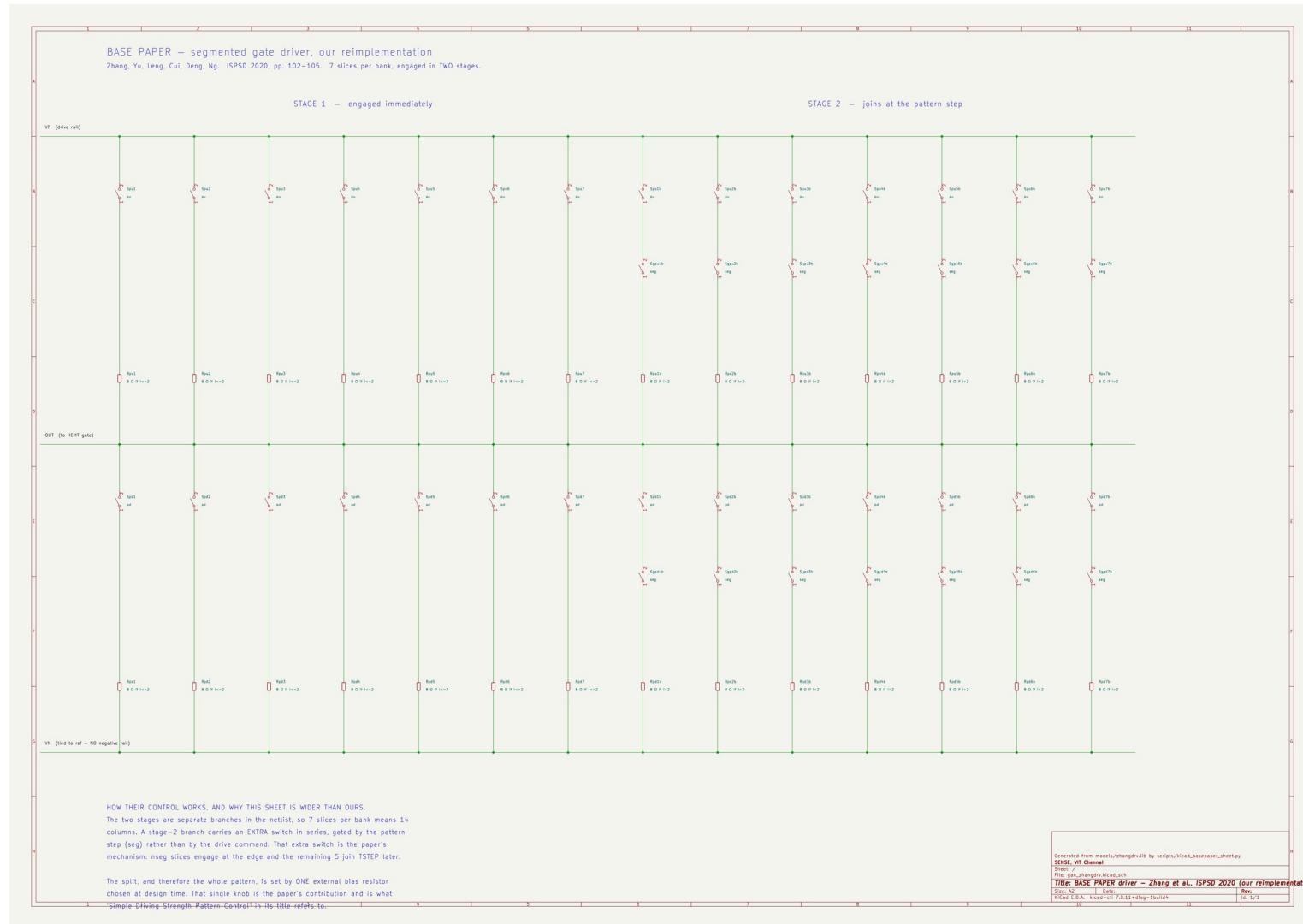


Fig. 9 - [kicad/gan_zhangdrv.kicad_sch](#), the base paper's driver reimplemented in the same testbench. Seven slices a bank in two stages, one bias resistor, no clamp branch, and the off rail tied to reference.

Segmented Driver: Block Structure

[kicad/gan_driver.drawio](#), the block diagram of [models/segdrv.lib](#).

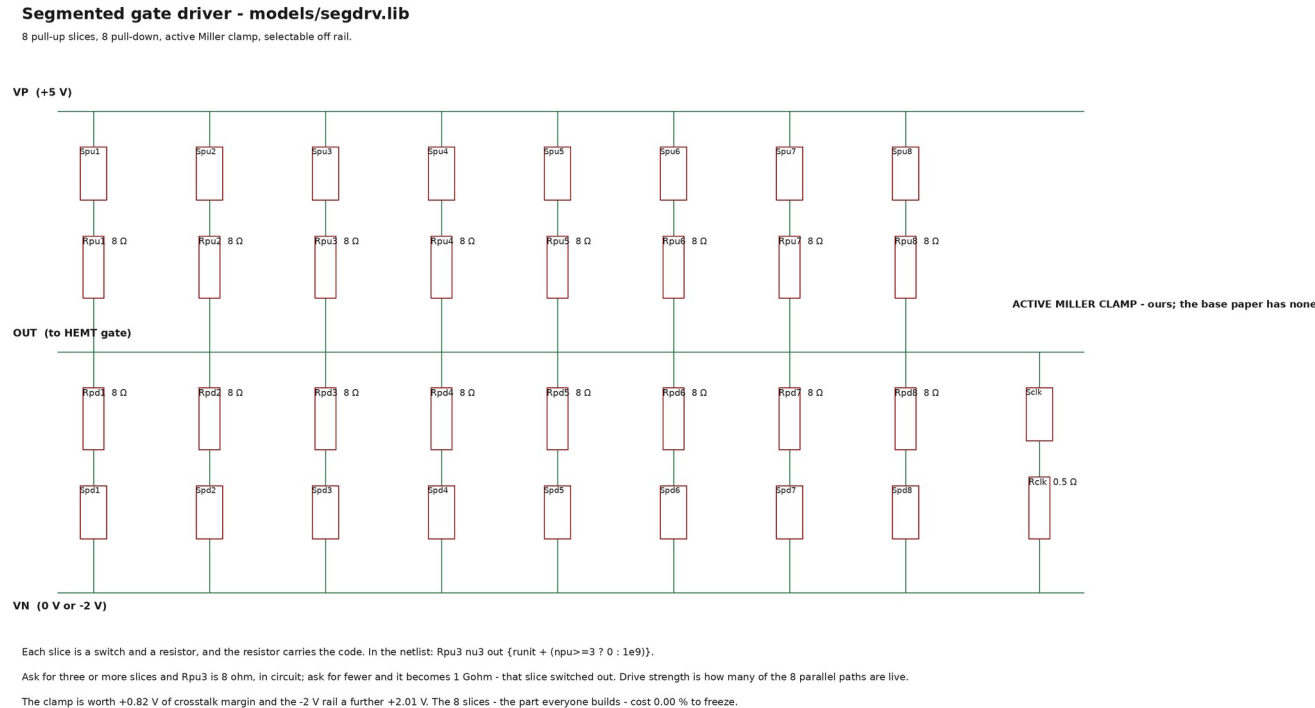


Fig. 10 - Sixteen slices between three rails: eight pull-up from VP, eight pull-down to VN, and OUT going to the HEMT gate. Each slice is one switch and one resistor, and the resistor carries the control word: ask for three or more slices and Rpu3 is 8 ohms and in circuit, ask for fewer and it becomes 1 G-ohm and that slice is out. Drive strength is simply how many of the eight parallel paths are live. On the right, the active Miller clamp, which the base paper does not have: its own switch and 0.5 ohms to VN, and VN itself selectable to -2 V.

Segmented Driver: SPICE Implementation

models/segdrv.lib - the output stage varied throughout the study.

models/segdrv.lib

the segmented output stage under study

```
23 .subckt SEGDRV pu pd clk out vp vn ref params: npu=8 npd=8 runit=8 rclamp=0.5  ◀ npu / npd are the 4 + 4 bits the FPGA emits
. . .
26 Spu1 vp nu1 pu ref SWP
27 Rpu1 nu1 out {runit + (npu>=1 ? 0 : 1e9)}  ◀ in circuit when npu ≥ 1, and 1 GΩ out of it when not
28 Spu2 vp nu2 pu ref SWP
. . .
43 * ---- pull-down bank : slice enabled if npd >= i -----
44 Spd1 nd1 vn pd ref SWP
45 Rpd1 out nd1 {runit + (npd>=1 ? 0 : 1e9)}
. . .
61 * ---- active Miller clamp : strong, separately timed -----
62 Sclk out nclk clk ref SWP  ◀ the clamp: its own switch, its own timing
63 Rclk nclk vn {rclamp}  ◀ 0.5 Ω to the off rail — where the injected charge goes
. . .
65 .ends SEGDRV
```

Fig. 11 - Eight pull-up slices and eight pull-down, each a switch and a resistor. A slice is in circuit when the control word reaches it and 1 GΩ out of it when it does not, which is how 4 + 4 bits from the FPGA become a drive strength. The clamp is not one of the slices: it has its own switch, its own timing and its own 0.5 Ω path to the off rail, because it has to hold the gate down while the other device switches. The slices are discrete by design: a single variable resistor would simulate identically and could not be laid out.

GaN HEMT Device Model

models/egan.lib - written from the EPC2010C datasheet.

models/egan.lib

the GaN HEMT model, written from the datasheet

```
19 .func smax(x) {0.5*(x+sqrt(x*x+1e-4))}  ◀ smooth max, so the solver does not stall at the kink
. . .
27 .model DGD D(IS=1e-30 N=40 CJO=150p VJ=1.0 M=0.65 FC=0.5) $ Crss (Miller)  ◀ C_GD — the crosstalk path, as a C(V) law
28 .model DDS D(IS=1e-30 N=40 CJO=800p VJ=1.0 M=0.5 FC=0.5) $ Coss-Crss
. . .
30 .subckt EGAN d g s params: vth=1.4 bh=5.55 rg=0.4 cgs=350p lg=0.3n ls=0.15n
. . .
42 Bch d si I = {bh} * ( smax(v(g2,si)-{vth})*smax(v(g2,si)-{vth})  ◀ one symmetric square law — reverse conduction follows from it
43 +                      - smax(v(g2,d) -{vth})*smax(v(g2,d) -{vth}) )
```

Fig. 12 - Vendor subcircuits are LTspice-dialect and do not port to Spectre, so this is written from datasheet quantities only. The channel is one symmetric square law, so third-quadrant conduction follows from the physics rather than being added: GaN has no body diode, and reverse conduction costs $V_{th} + |V_{off}| + I \cdot R_{ds(on)}$. That is the coupling the project turns on - a negative off rail improves crosstalk margin at the cost of dead-time loss. C_{GD} is a junction diode biased never to conduct, so only its $C(V)$ law is used. The temperature coefficients are hand-typed from the datasheet, not fitted; this is the principal limitation of the model, and it is covered in the backup deck.

Methodology

Every comparison is the same file with one thing changed. The following is held identical; the listed parameter is the only one that varies.

HELD IDENTICAL

the netlist [sim/dpt.cir](#), byte for byte - not a copy, not a variant

the device [models/egan.lib](#) on both sides, same V_{th} , same $R_{es(on)}$, same C_{9e}

the parasitics 3 nH power loop, 0.3 Ω loop resistance, same gate inductance

the operating point 100 V bus, 10 A load, 25 °C junction

the solver same timestep, same options, same initial conditions

ENVIRONMENT ngspice-42 and KiCad 7.0.11 on a Debian container, headless, no hardware in the loop. Both are open source; there is no licensed tool and no vendor model anywhere in this project.

THE ONE THING THAT MOVES

theirs vs ours the driver subcircuit, and nothing else:
[models/zhangdrv.lib](#) ↔ [models/segdrv.lib](#)

GaN vs silicon the device model only, $R_{es(on)}$ matched at the 25 m Ω class

clamp on/off one parameter, CLKEN; off rail, one parameter, VNEG

Fig. 13 - Their driver is additionally given a freedom its own paper does not have: [scripts/headtohead.py](#) searches both of its controls at every corner and it is run at the setting that search finds optimal for it, read out of [results/headtohead.txt](#). Ours runs one fixed control word at every corner. Any remaining difference is the driver. [results/RESULTS-SUMMARY.txt](#) names the script behind each number, and [review/check_consistency.py](#) fails the build if a slide and the data disagree.

Base Paper Driver: Simulation Result

[models/zhangdrv.lib](#) on [sim/dpt.cir](#). Seven slices per bank in two stages, no clamp branch, no negative off rail.

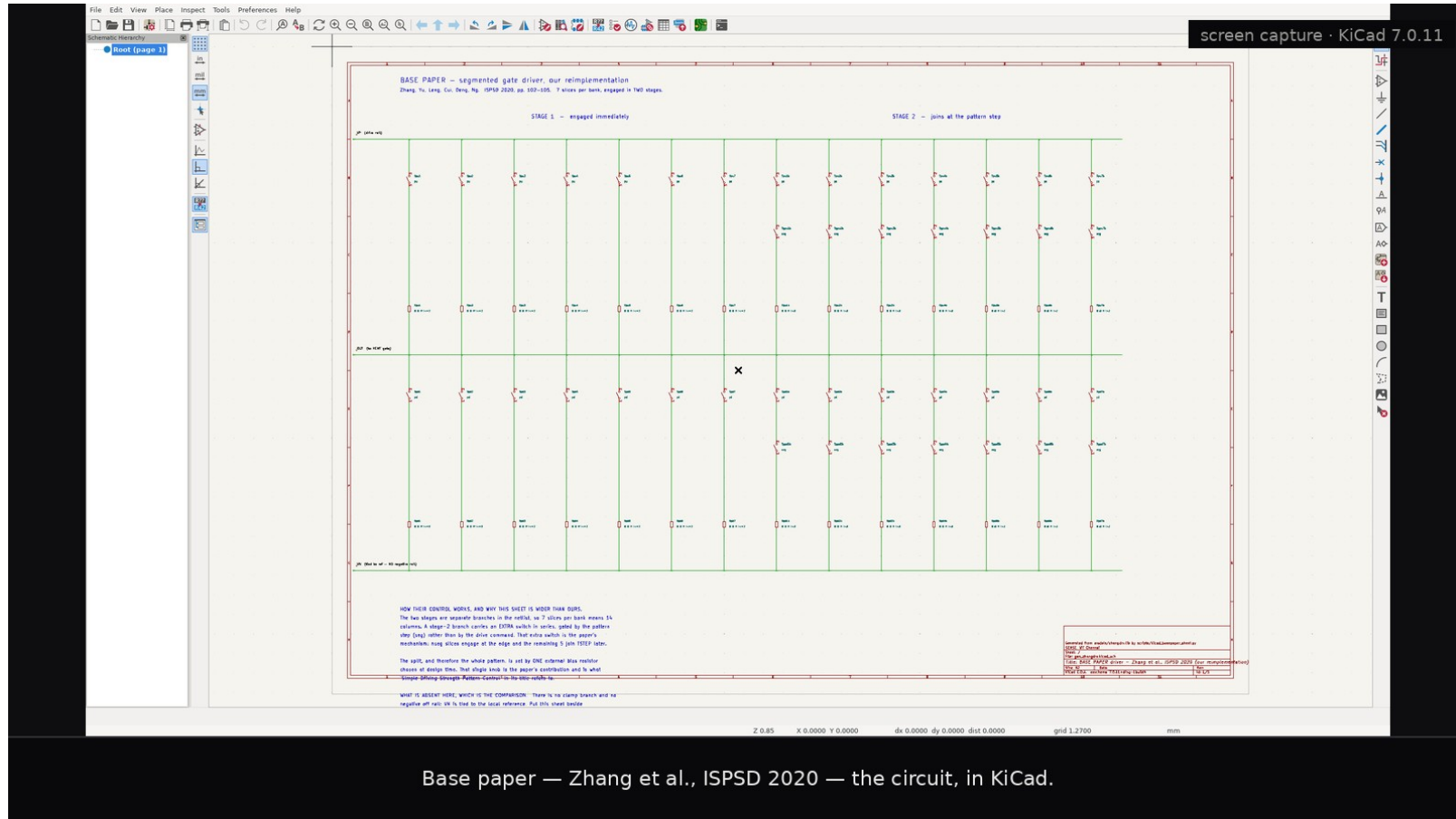
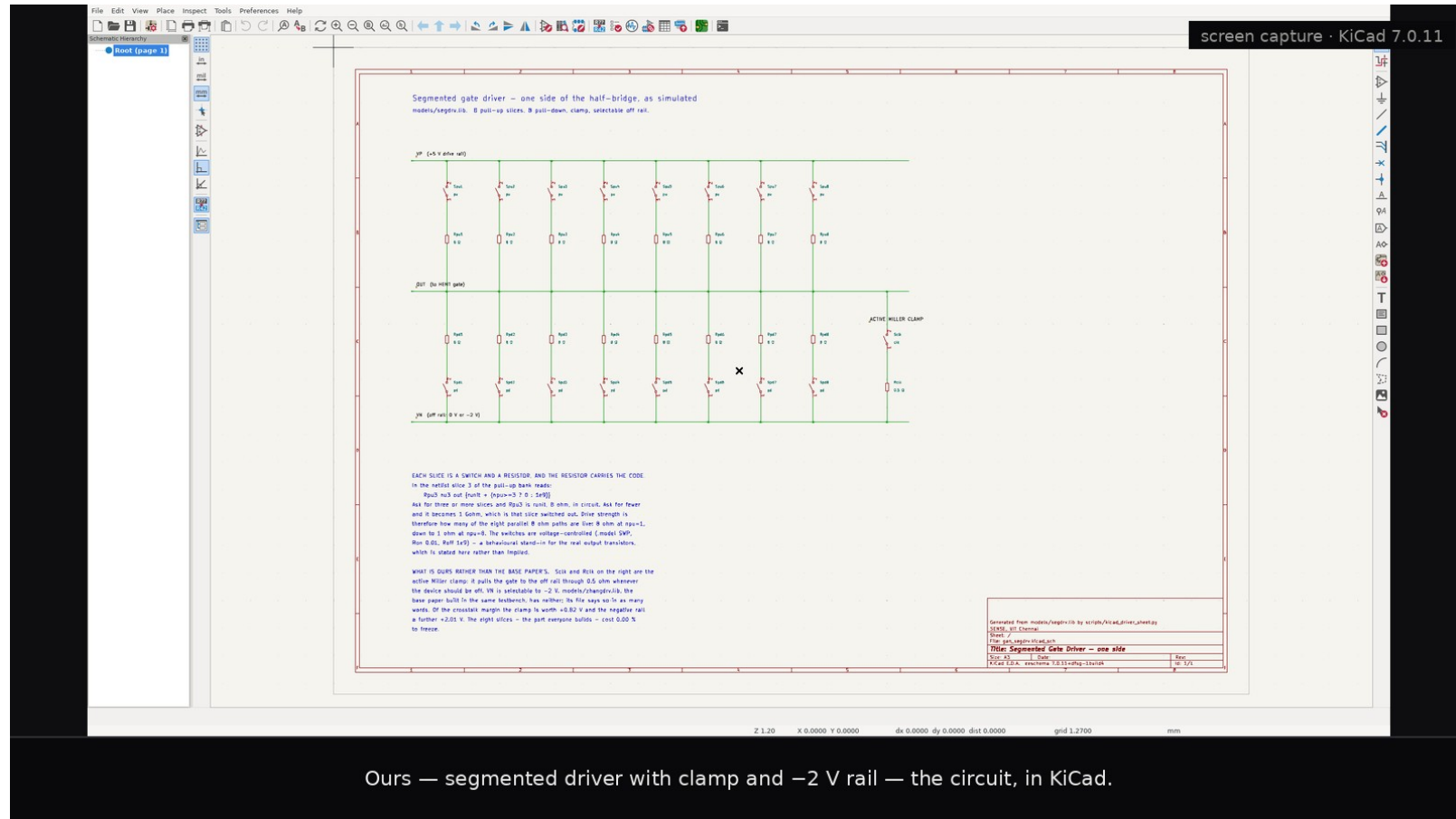


Fig. 14 - Their circuit in KiCad ([kicad/gan_zhangdrv.kicad_sch](#)), ngspice running it, and the waveform it produced. The OFF device's gate peaks at +0.993 V against a 1.400 V threshold - 0.407 V of margin. Run at $n_{\text{seg}}=2$, $t_{\text{step}}=4n$, the setting [scripts/headtohead.py](#) found optimal for their driver at this corner, read from [results/headtohead.txt](#): their own paper sets one bias resistor once at design time and gets no per-corner choice at all. 35 s.

Proposed Driver: Simulation Result

[models/segdrv.lib](#) on the same [sim/dpt.cir](#). Eight slices per bank, active Miller clamp, -2 V off rail.



Ours — segmented driver with clamp and -2 V rail — the circuit, in KiCad.

Fig. 15 - The same sequence over our driver. The gate peaks at $-1.176\text{ V} - 2.576\text{ V}$ of margin, $6.3\times$ theirs. One fixed control word, the same one used at every corner in the study, not re-tuned for this run. The plot axes are identical to the previous slide's and fixed in the script, so the two are directly comparable. 35 s.

Demonstration

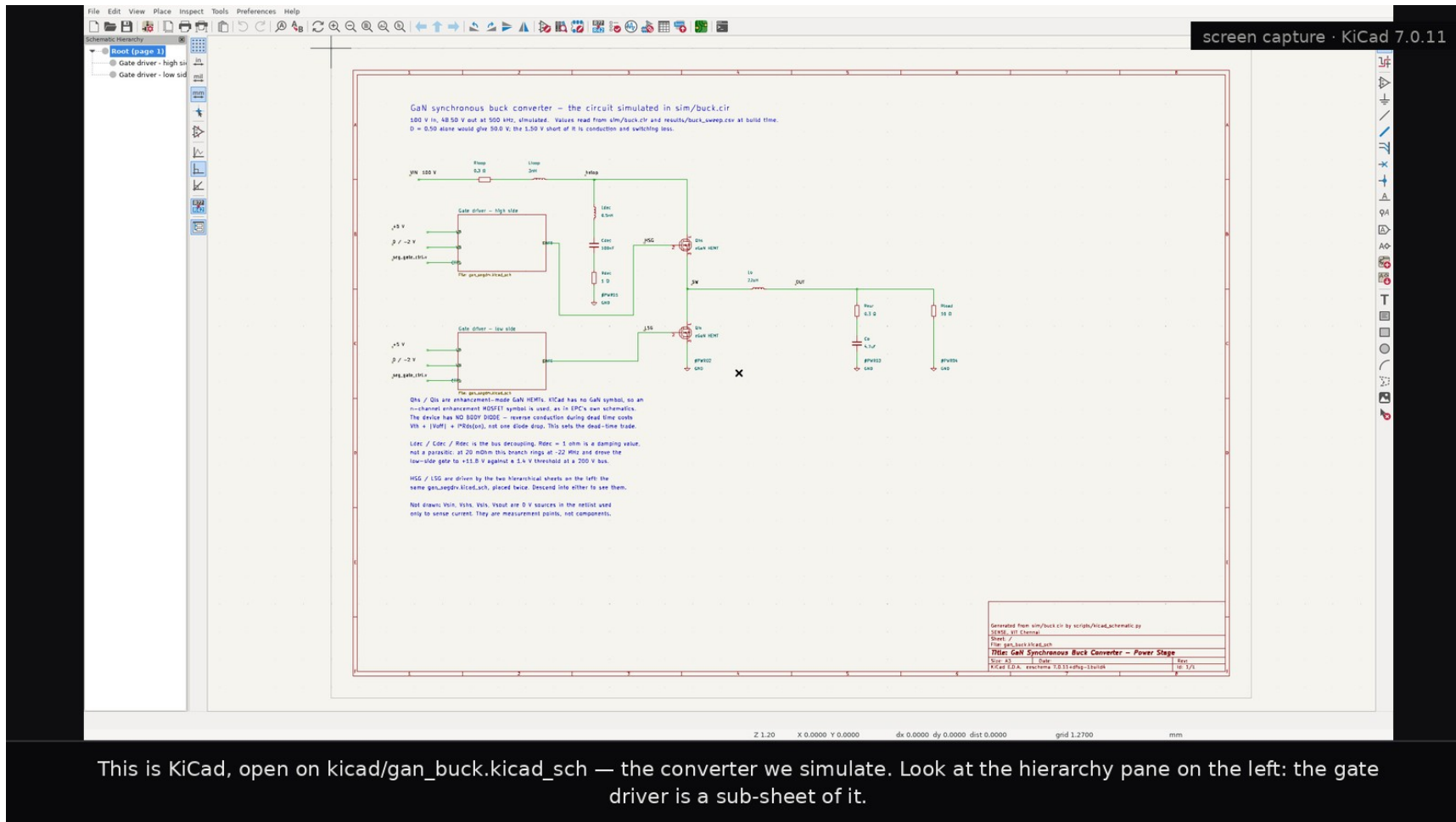


Fig. 16 - Four sections, 100 s. The circuit as its KiCad sheet ([kicad/gan_buck.kicad_sch](#)), the tools named and versioned, the output with the threshold, the peak and the margin annotated, and the base paper's driver run against ours on the same bench at 100 V / 10 A / 25 °C. Theirs peaks +0.993 V, 0.407 V short of the 1.400 V threshold; ours peaks -1.176 V, 2.576 V clear. The converter it drives: 48.50 V out at 236.26 W, 97.49 % efficient. Three ngspice runs made while the film was building - rebuilt by [scripts/demo_review2.py](#), so every number on screen is computed, not captioned.

Live Simulation

bash [proof/LIVE-SIM.sh](#) - two ngspice transients, about ten seconds, on the presentation machine.

```
$ bash proof/LIVE-SIM.sh 5.0 s

GaN half-bridge -- crosstalk, simulated live
-----
machine  vm      2026-09-27 13:24:17
spice    ** ngspice-42 : Circuit level simulation program
circuit  sim/dpt.cir      device  models/egan.lib
changing ONLY CLKEN (clamp) and VNEG (off rail)

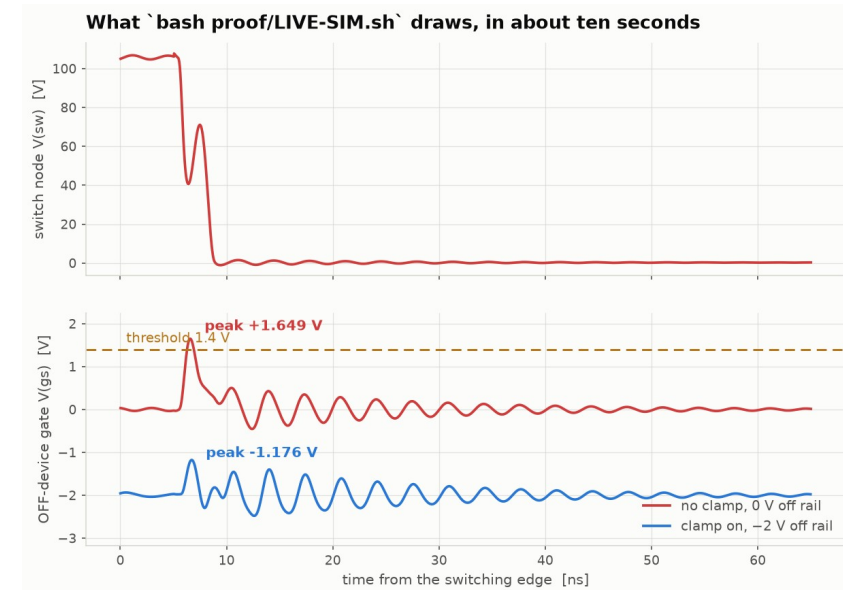
(a) fastest drive, no clamp, 0 V off rail ...
    OFF-device gate reaches +1.6488 V  against a 1.400 V threshold
    margin -0.2488 V    false_turn_on = 1  SHOOT-THROUGH

(b) same circuit, clamp on, -2 V off rail ...
    OFF-device gate reaches -1.1759 V
    margin +2.5759 V    false_turn_on = 0  SAFE

-----
                                on the slide    this run
OFF gate, no clamp [V]          1.65           1.6488
margin, shipped [V]             2.576          2.5759
-----
two ngspice transients, 3.7 s wall clock on this machine

drawing the waveforms those two runs produced ...
```

recorded run of the same command



the waveforms it produces

Fig. 17 - Simulated in ngspice. Two runs of one circuit file. Between them only CLKEN (the Miller clamp) and VNEG (the gate off rail) change, so no other parameter varies. It prints what it measured beside what these slides claim - OFF gate 1.649 V against the deck's 1.65, margin 2.576 V against the deck's 2.576 - then draws the waveforms those two runs produced. 3.8 s of that is ngspice; the rest is drawing. [scripts/live_demo.py](#).

Crosstalk: Fault and Mitigation

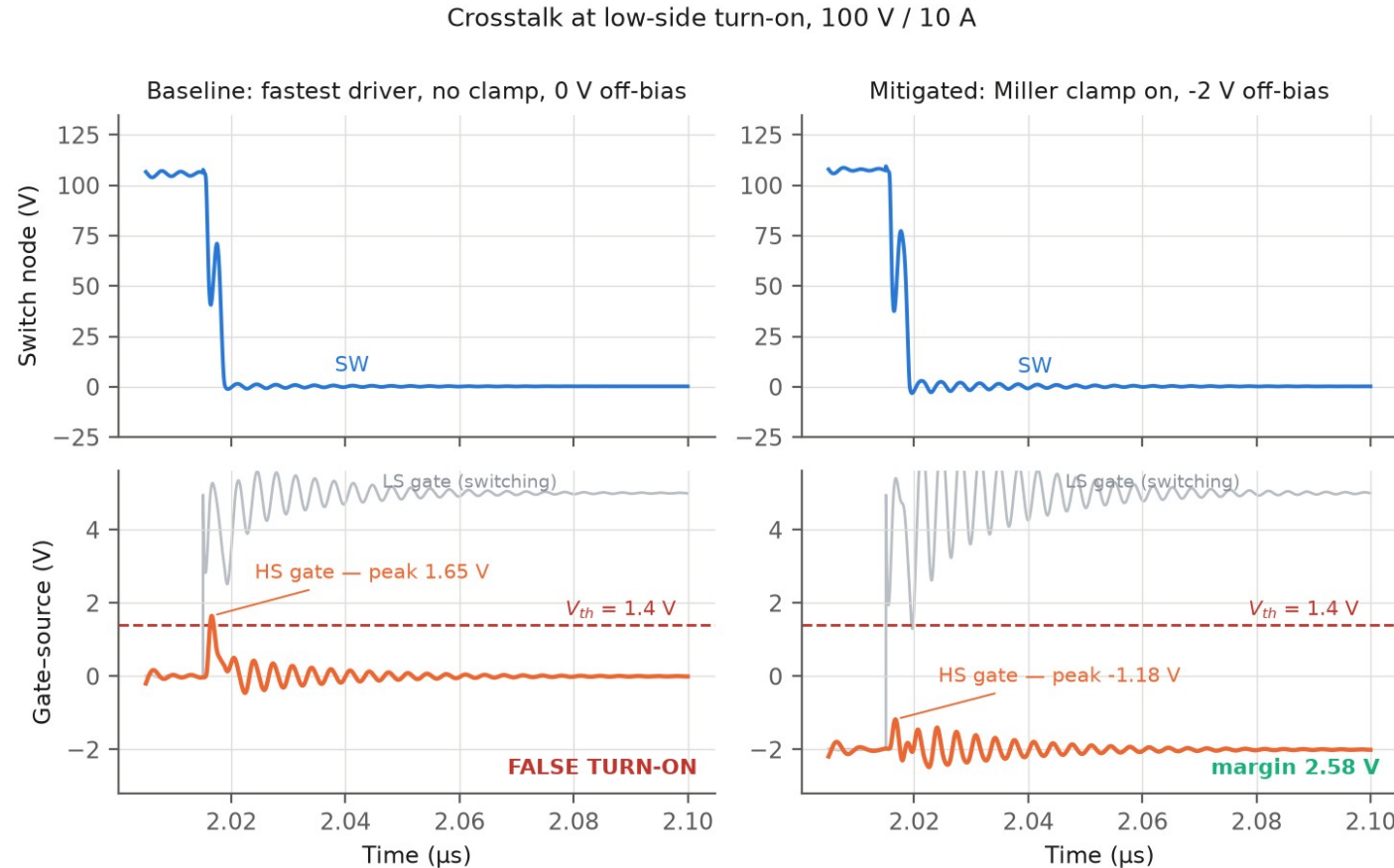


Fig. 18 - Simulated in ngspice. Two runs of [sim/dpt.cir](#). Top row: the switch node falls from 106 V in 2.90 ns, peaking at 124 V/ns - that dv/dt drives $i = C_{GD} \cdot dv/dt$ into the OFF device's gate, and is the cause. Bottom row: that gate. Left, without the clamp, it rests at +0.005 V, is lifted +1.643 V and peaks at +1.649 V, past the 1.4 V threshold - false_turn_on = 1. Right, with the clamp and the -2 V rail, it rests at -1.941 V, is lifted only +0.765 V and peaks at -1.176 V - false_turn_on = 0. The two fixes are not one mechanism: the rail moves where the gate starts, the clamp shortens the lift by giving the injected charge a 0.5 Ω path out. [scripts/waveform_anatomy.py](#).

Six-Parameter Comparison

Column 1 → 2 changes only the device. Column 2 → 3 changes only the control.

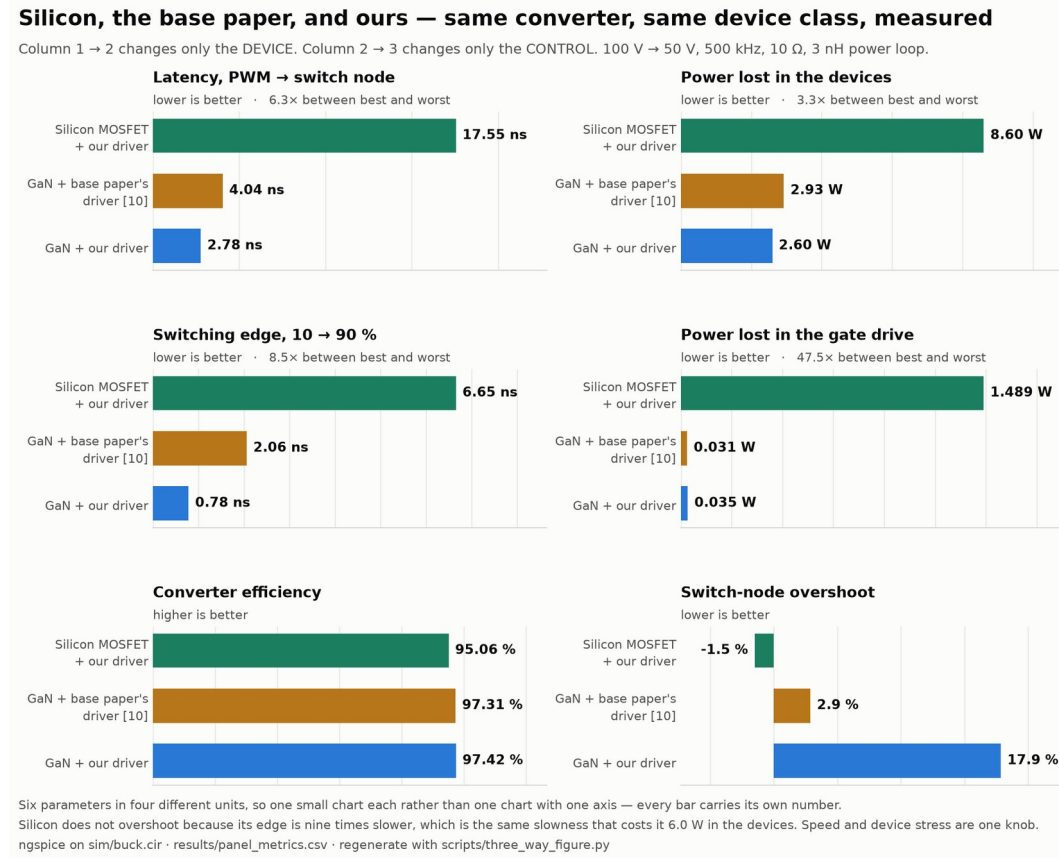


Fig. 19 - Simulated in ngspice. Latency 17.55 → 4.04 → 2.78 ns and device power 8.60 → 2.93 → 2.60 W, left to right. Against silicon that is 6.3× and 3.3×; against the base paper's control on the same GaN device, 1.5× and 1.1×. The device swap accounts for most of it; the control swap accounts for the rest. Overshoot is the row that runs the other way, and it runs that way for the same reason the other five do not: silicon does not overshoot because its edge is nine times slower, and that slowness is what costs it 6.0 W. Speed and device stress are one knob.
[scripts/three_way_figure.py](#), from [results/panel_metrics.csv](#).

Comparison with the Base Paper

Same netlist, same GaN, same power loop, same parasitics - only the driver is swapped. We hold one fixed word at all four corners. They are re-optimised at every corner, which is more freedom than their own design has.

crosstalk margin	base, as their paper builds it	base, re-tuned at every corner	ours, one fixed word	vs their best
50 V / 2 A / 25 °C	+0.503 V	+0.503 V	+2.757 V	5.5×
100 V / 10 A / 25 °C	+0.407 V	+0.407 V	+2.576 V	6.3×
200 V / 2 A / 125 °C	+0.261 V	+0.261 V	+2.309 V	8.9×
200 V / 10 A / 125 °C	+0.181 V	+0.181 V	+2.251 V	12.4×

The margin gap widens with stress

Their margin falls +0.503 → +0.181 V from the mildest corner to the hottest. Ours goes +2.757 → +2.251. They degrade where it matters most; a clamp does not care how hot the device is.

Worst corner is what a converter has to survive: +0.181 V against +2.251 V.

The edge is faster, not slower

Their scheme reduces crosstalk by slowing the edge: 67–103 V/ns at the switch node. Ours runs 101–175 V/ns - about twice as fast - and still wins by a factor of several.

The margin is not bought with switching speed. That is the useful form of the result.

What it costs, and what we are not claiming. Our turn-on energy is higher at three of four corners - the –2 V rail deepens GaN's dead-time reverse drop, a penalty this project already measures at 1.0–3.4 % of total loss. Their driver needs one bias resistor; ours needs a clamp device, a negative supply and 20 LUTs. For a converter that never leaves one operating point, theirs may be the right engineering.

This is our implementation of their described scheme, run in our testbench: their netlist is not published. The ratio is against that reimplementations, not against their measured result, and is not claimed as such.

Next Steps

What remains is bench work, not further simulation.

Place-and-route on a chosen board. Synthesis is done; the flow stops there because the XDC pins are placeholders. Doing it properly also means driving the 200 MHz clock from an MMCM rather than straight off a pin, which is what makes 34 clock-to-pin paths fail today.

A hardware half-bridge, measured. This is the whole of the remaining risk. Everything in this deck is a simulation of a converter that has never been built, and one behavioural GaN model underlies all of it.

Additionally: transcribe the silicon MOSFET datasheet digits rather than using datasheet-class values, and re-run the ceiling on the transistor-level stage now that it is known to work.

References (1-15)

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- [2] R. Xie, H. Wang, G. Tang, X. Yang and K. J. Chen, "An Analytical Model for False Turn-On Evaluation of High-Voltage Enhancement-Mode GaN Transistor in Bridge-Leg Configuration," IEEE Trans. Power Electron., vol. 32, no. 8, pp. 6416-6433, Aug. 2017.
- [3] D. Reusch and J. Strydom, "Understanding the Effect of PCB Layout on Circuit Performance in a High-Frequency Gallium-Nitride-Based Point of Load Converter," IEEE Trans. Power Electron., vol. 29, no. 4, pp. 2008-2015, Apr. 2014.
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- [5] X. Chen, H. Peng, S. Song, Q. Tong and Y. Kang, "A Novel Control Strategy for Optimal Tradeoff between Overshoot and Switching Loss Based on Double Closed-Loop Self-Regulating Active Gate Driver," IEEE Trans. Power Electron., vol. 39, no. 10, pp. 13033-13043, 2024.
- [6] W. Song, T. Hu, J. Chen, T. Tang, H. Yue and G. Liu, "A Self-Regulating Active Gate Driver of Voltage Overshoot Suppression for SiC MOSFETs Under Variable Load Current Conditions," IEEE Trans. Power Electron., vol. 40, no. 8, pp. 10623-10634, 2025.
- [7] C.W. Kuo, T.W. Wang, L.C. Chen, C.C. Tu, Y.K. Hsiao and P.H. Chen, "Universal Active Gate Driver IC With Closed-Loop Timing Control and Gate-Sensing Technique for Silicon Carbide Power Devices," IEEE Trans. Power Electron., vol. 40, no. 4, pp. 5120-5129, 2025.
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The papers this design engages with directly. [10] is the base paper, reimplemented in [models/zhangdrv.lib](#) and measured against throughout.

Thank you

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Everything in this deck runs on request: github.com/Amritha902/gan-driver